SHIFTGUARD

Two-dataset evaluation report

Explainable and Uncertainty-Aware Intrusion Detection under Network Distribution Shift

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10 October 2026 (Bishkek) | Actual uploaded CSV data | One development seed (42)

What the measured results show

UNSW-NB15: the validation-selected XGBoost model achieves test macro-F1 0.8919 and attack recall 86.36%. Its frozen review policy refers 17.37% of test flows, exceeding the 15% source budget.

CICIDS2017: under the Friday holdout, selected XGBoost falls from validation macro-F1 0.9983 to test macro-F1 0.3779. Attack recall is only 0.68%. It automatically misses 92.79% of true attacks after the uncertainty gate.

These findings demonstrate a failure mode of source-trained confidence-based referral. They do not support operational reliability or successful zero-day detection. Low false-positive rates on CIC coexist with extremely poor attack recall.

Dataset	Fit rows	Calibration	Validation	Test rows
UNSW-NB15	44,322	14,945	14,744	174,401
CICIDS2017	111,344	38,474	37,067	702,694

Two independent feature schemas were fitted. No UNSW-to-CIC transfer was run. The dashboard displays a 5,000-row uniform test sample for each dataset; this report uses full cleaned held-out-test evaluations.

1. Dataset inventory and audit

archive (4).zip contains four headerless UNSW raw CSVs, a feature dictionary, event metadata, and two curated partitions. The raw files are inventoried separately and are not pooled with the overlapping curated subsets.

UNSW file	Rows	Role
UNSW-NB15_1.csv	700,001	Raw inventory
UNSW-NB15_2.csv	700,001	Raw inventory
UNSW-NB15_3.csv	700,001	Raw inventory
UNSW-NB15_4.csv	440,044	Raw inventory
UNSW_NB15_training-set.csv	82,332	development source
UNSW_NB15_testing-set.csv	175,341	held-out test

Supplied training has 82,332 rows and testing has 175,341; opposite to the conventional published sizes. Filenames are preserved, not silently swapped.

archive (3).zip contains eight CIC flow CSVs. Their capture-day filenames are available; timestamps, endpoint addresses and session IDs are absent from these machine-learning CSVs.

CIC file	Raw rows	Valid rows
Friday-WorkingHours-Afternoon-DDos.pcap_ISCX.csv	225,745	225,637
Friday-WorkingHours-Afternoon-PortScan.pcap_ISCX.csv	286,467	286,291
Friday-WorkingHours-Morning.pcap_ISCX.csv	191,033	190,808
Monday-WorkingHours.pcap_ISCX.csv	529,918	529,383
Thursday-WorkingHours-Afternoon-Infilteration.pcap_ISCX.csv	288,602	288,249
Thursday-WorkingHours-Morning-WebAttacks.pcap_ISCX.csv	170,366	170,103
Tuesday-WorkingHours.pcap_ISCX.csv	445,909	445,393
Wednesday-workingHours.pcap_ISCX.csv	692,703	691,953

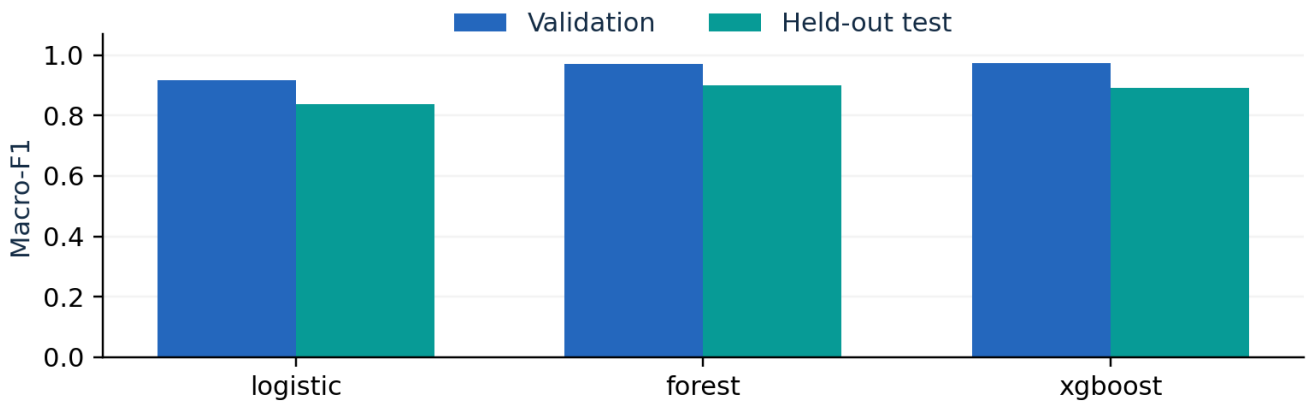
CIC total: 2,830,743 raw rows; 2,827,817 remain after measurement/label cleaning, before leakage and conflict exclusions. Raw archive member counts are reported as observed, without forcing published counts.

2. UNSW-NB15 measured results

Supplied filename direction preserved: 82,332 development-source rows and 175,341 test-source rows. The cleaned development source is split by indivisible feature signatures into fit/calibration/validation partitions.

Model	Validation macro-F1	Test macro-F1	Attack recall	FPR	Attack AP
logistic	0.9181	0.8386	79.12%	3.33%	0.9868
forest	0.9701	0.8998	87.71%	2.58%	0.9936
xgboost *	0.9730	0.8919	86.36%	2.16%	0.9930

* Selected by validation macro-F1. Test outcomes never determine selection.



Selected XGBoost policy	Measured value
Validation-selected threshold	0.9800
Validation review fraction	14.68%
Full-test review fraction	17.37%
Automatically missed attacks	4,774 / 118,797
Auto-missed attack rate	4.02%

Random Forest has a higher test macro-F1 than XGBoost in this run, but the application retains the validation-selected XGBoost. Choosing the best test performer after observing these results would contaminate model selection.

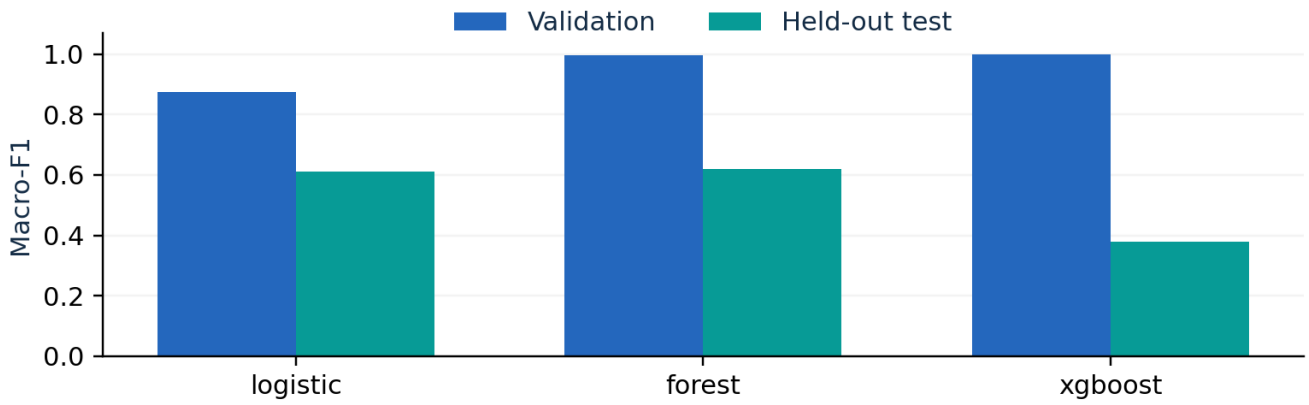
Audit exclusions: 8,313 development rows match test signatures; 14 development rows and 940 test rows participate in binary-label-conflicting signatures. Exclusion categories can overlap; the retained total is reported independently.

3. CICIDS2017 Friday shift results

Development uses a uniform, seed-42 reservoir capped at 200,000 valid Monday-Thursday rows. After duplicate/conflict auditing, 186,885 sampled rows remain. Friday rows retained after measurement cleaning and detected label-conflict exclusions are held out. Destination Port and the duplicate header-length column are excluded from the 76 predictors.

Model	Validation macro-F1	Test macro-F1	Attack recall	FPR	Attack AP
logistic	0.8742	0.6120	28.09%	1.44%	0.8481
forest	0.9970	0.6193	28.18%	0.07%	0.8746
xgboost *	0.9983	0.3779	0.68%	0.04%	0.8070

* Selected by validation macro-F1. Test outcomes never determine selection.



Selected XGBoost policy	Measured value
Validation-selected threshold	0.9900
Full-test review fraction	2.98%
Automatically missed attacks	268,041 / 288,870
Auto-missed attack rate	92.79%
Accepted prediction error	39.33%

Friday contains Bot, DDoS and PortScan categories absent from the Monday-Thursday source categories. This is a binary unseen-attack-type/day-shift stress test, not proof of real-world zero-day detection. The near-perfect source validation score did not transfer.

The low review fraction is not a success: most attacks are confidently classified as Benign. Additional OOD/referral strategies and independent target-labelled evaluation are needed before practical use. The held-out Friday data have not been used to tune or replace the selected model.

4. Binary detection by attack category

The models predict Attack or Benign. The following category names are ground-truth strata used for evaluation, not predicted incident categories. Correct binary fraction is attack recall for malicious rows and specificity for benign rows.

UNSW-NB15

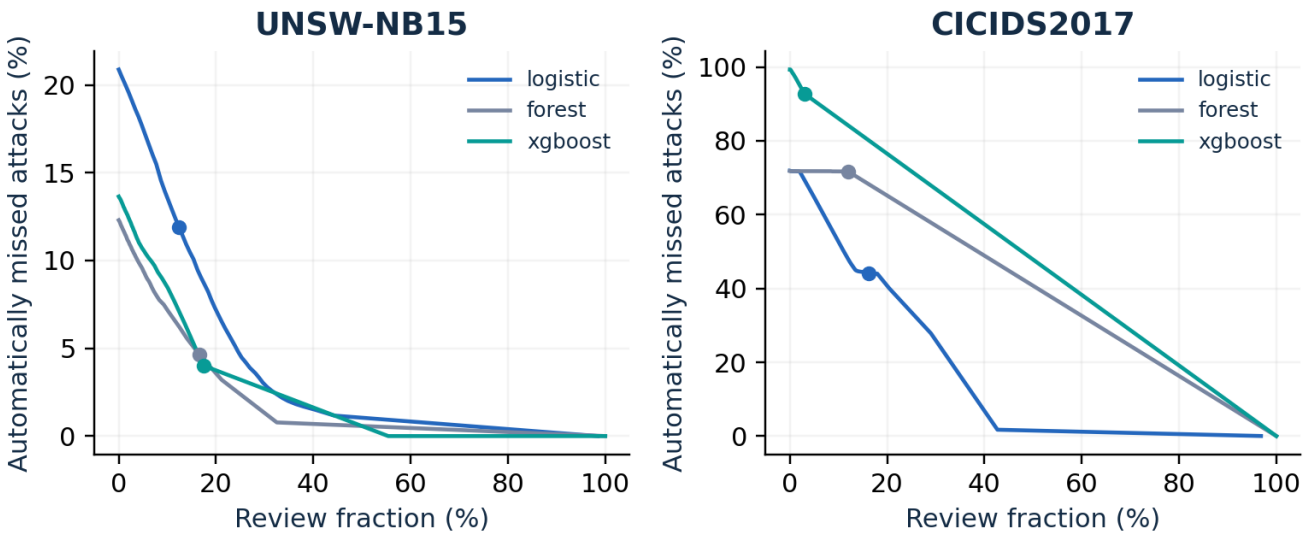
Ground-truth category	Test rows	Correct binary	Review	Auto misses
Analysis	2,000	71.20%	26.65%	399
Backdoor	1,746	97.65%	21.53%	4
DoS	12,264	96.25%	25.16%	40
Exploits	33,393	95.55%	18.18%	177
Fuzzers	17,641	25.92%	65.98%	4,124
Generic	40,000	99.87%	0.60%	9
Normal	55,604	97.84%	11.23%	0
Reconnaissance	10,491	96.88%	13.81%	11
Shellcode	1,132	82.77%	56.63%	10
Worms	130	99.23%	14.62%	0

CICIDS2017

Ground-truth category	Test rows	Correct binary	Review	Auto misses
Benign	413,824	99.96%	0.06%	0
Bot	1,933	0.00%	0.00%	1,933
DDoS	128,007	1.38%	15.98%	107,550
PortScan	158,930	0.12%	0.17%	158,558

UNSW Fuzzers are particularly weak despite strong aggregate attack recall. In the CIC Friday holdout, Bot detection recall is zero and every Bot flow is automatically missed. Aggregate accuracy alone would conceal these failures.

5. Uncertainty and review workload

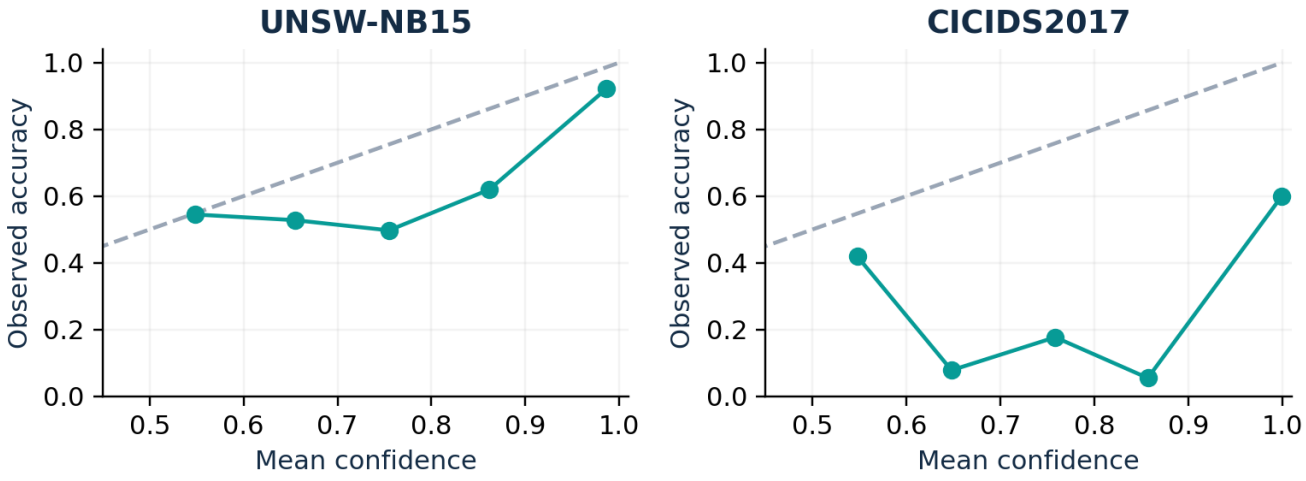


Each dot marks the frozen threshold selected on source validation with a review-budget ceiling of 15%. Test curves are diagnostic; their better-looking points must not be used to retune the reported threshold.

Automatic missed-attack rate counts true attacks classified Benign outside the review queue, divided by all true attacks. Referred flows are unresolved. No perfect analyst or successful recovery is assumed.

Selected model	Review count	Attention count	Attention fraction
UNSW-NB15	30,299	120,602	69.15%
CICIDS2017	20,963	21,125	3.01%

Attention combines predicted-attack alerts and uncertainty referrals, counting their overlap once. It measures flow-level demand, not unique incidents or analyst minutes. Incident aggregation, reviewer time and backlog remain pilot measurements.



Selected-model 10-bin ECE: UNSW 0.0723; CIC 0.4036. Confidence calibration on source data does not guarantee target calibration. Reliability bins are empirical summaries, not confidence intervals.

6. Input shift and explanation evidence

Input monitoring compares the fit-partition reference with the full held-out test. Numeric PSI uses training-quantile bins; categorical total variation uses the training vocabulary. Missing-value fractions are reported separately. Shift scores do not establish incident causation or confirm model failure by themselves.

UNSW-NB15 - five largest feature shift scores

Feature	Metric	Shift score	Test missing
sinpkt	PSI	0.2269	0.00%
dur	PSI	0.1318	0.00%
proto	Total variation	0.1272	0.00%
state	Total variation	0.1157	0.00%
sbytes	PSI	0.1098	0.00%

CICIDS2017 - five largest feature shift scores

Feature	Metric	Shift score	Test missing
Init_Win_bytes_backward	PSI	1.4759	40.37%
Max Packet Length	PSI	0.5346	0.00%
Bwd Header Length	PSI	0.4401	0.00%
Fwd Packet Length Max	PSI	0.4238	0.00%
Avg Fwd Segment Size	PSI	0.3959	0.00%

The explanation method replaces one feature with its fit-partition median/mode, then measures the change in the predicted-class probability. It is neither SHAP nor an additive decomposition. Correlated-feature replacements can be unrealistic.

Recorded confidently wrong CIC example

Flow: CIC:Friday-WorkingHours-Afternoon-DDos.pcap_ISCX.csv:90354; ground truth: DDoS; prediction: Benign; confidence: 99.98%. The explanation describes why the model favours its output, not whether that output is correct.

Feature	Observed	Training reference	Delta p
Bwd Header Length	0.0	40.0	+0.00011
Min Packet Length	6.0	0.0	+0.00010
min_seg_size_forward	20.0	32.0	+0.00008
Init_Win_bytes_forward	256.0	1024.0	-0.00001

7. Reproducibility, limits and next experiments

The updated application contains six trained baseline models, an added Dataset reports screen, per-dataset audit/evaluation JSONs, baseline comparison CSVs, review curves, reliability summaries, real 5,000-row held-out sample CSVs and their prediction exports. Synthetic examples remain explicitly marked.

Raw source archives and full prepared partitions are not duplicated in the application ZIP. To reproduce, place the supplied archives at known local paths and run the dataset project command with --unsw-zip and --cic-zip, or extract their CSVs into the documented raw folders. The command recreates audits, partitions, fitted models and reports.

CIC cleaning: trim names; verify the duplicated Fwd Header Length column; exclude Destination Port; convert infinities and -1 initial-window sentinels to missing; drop other negative measurements and missing labels; fit medians/scales only on fit rows. Garbled web-attack label names are normalized for display. No existing CSV columns are treated as a validated UNSW/CIC common schema.

CIC leakage audit: 138,119 valid development-pool rows match Friday signatures; 12,972 sampled rows are removed for that reason. Binary conflicts affect 5,136 full development-pool rows, 514 sampled rows and 42 test rows. These reason counts overlap; retained totals are authoritative.

Limitations: one seed; no group-bootstrap confidence intervals; development subsampling for CIC; conservative feature-signature conflict exclusions can alter the evaluated population; no endpoint/session split in curated CSVs; no cross-dataset transfer; no authorized campus pilot; no live capture or automatic blocking. Timing is one predict_proba batch, excluding preprocessing/explanations, not an end-to-end p95 service measurement.

Next experiments: compare raw versus calibrated referral, matched-workload random referral, entropy and independently validated OOD gates; preserve untouched target evaluation windows; test compatible PCAP-derived features from both datasets; run repeated seeds and group/time-block intervals; then conduct a scoped observation-only campus study with independent review and sampling of automatically accepted benign flows.

Operational conclusion: the CIC-selected model is unsuitable for dependable attack detection in the tested Friday setting. The implemented uncertainty gate does not correct the failure. The result is valuable as a transparent research stress test, not a deployment success claim.

Source provenance: filenames and member SHA-256 hashes appear in each dataset_report.json; processed-partition hashes and software versions accompany the evaluation. requirements-lock.txt records tested top-level versions. Full reports contain calibration ablations, confusion matrices, per-class scores, drift and timing.

Primary dataset access and attribution: https://research.unsw.edu.au/projects/unsw-nb15-dataset and https://www.unb.ca/cic/datasets/ids-2017.html. Dataset facts in this report were measured from the supplied archives.